

Python Assignment-19

1. Write a program which contains one lambda function which accepts one parameter and return power of two.

Input : 4 Output : 16

Input : 6 Output : 64

Answer :

```
Power = lambda iValue : 2 ** iValue

def main():
    Value = int(input("Enter the Number : "))

    Result = Power(Value)

    print(Result)

if __name__ == "__main__":
    main()
```

2. Write a program which contains one lambda function which accepts two parameters and return its multiplication.

Input : 4 3 Output : 12

Input : 6 3 Output : 18

Answer :

```
Multiplication = lambda iValue1,iValue2 : iValue1 * iValue2

def main():
    print("Enter the Number1")
    Value1 = int(input())

    print("Enter the Number2")
    Value2 = int(input())

    Result = Multiplication(Value1,Value2)
    print("Your Multiplication is : ", Result)

if __name__ == "__main__":
    main()
```

3. Write a program which contains filter(), map() and reduce() in it. Python application which contains one list of numbers. List contains the numbers which are accepted from user. Filter should filter out all such numbers which are greater than or equal to 70 and less than or equal to 90. Map function will increase each number by 10. Reduce will return product of all that numbers.

Input List = [4, 34, 36, 76, 68, 24, 89, 23, 86, 90, 45, 70]

List after filter = [76, 89, 86, 90, 70]

List after map = [86, 99, 96, 100, 80]

Output of reduce = 6538752000

Answer :

```
from functools import reduce

Filtering = lambda iValues : 70 <= iValues <= 90

Increase = lambda ifValues : ifValues + 10

Product = lambda imValue1, imValue2 : imValue1 * imValue2

def main():
    print("Enter the list of numbers")
    Values = list(map(int, input().split()))
    print("Input List : ", Values)

    fValues = list(filter(Filtering, Values))
    print("List After Filter : ", fValues)

    mValues = list(map(Increase, fValues))
    print("List After Map ", mValues)

    rValues = reduce(Product, mValues)
    print("Output of Reduce : ", rValues)

if __name__ == "__main__":
    main()
```

4. Write a program which contains filter(), map() and reduce() in it. Python application which contains one list of numbers. List contains the numbers which are accepted from user. Filter should filter out all such numbers which are even. Map function will calculate its square. Reduce will return addition of all that numbers.

Input List = [5, 2, 3, 4, 3, 4, 1, 2, 8, 10]

List after filter = [2, 4, 4, 2, 8, 10]

List after map = [4, 16, 16, 4, 64, 100]

Output of reduce = 204

Answer:

```

from functools import reduce

Even = lambda iValues : iValues%2==0

Square = lambda iValues : iValues * iValues

FinalOutput = lambda iValues1, iValues2 : iValues1 + iValues2

def main():
    print("Enter the list of numbers")
    Values = list(map(int,input().split()))

    fValues = list(filter(Even,Values))
    print("List After Filter", fValues)

    mValues = list(map(Square,fValues))
    print("List after map", mValues)

    rValues = reduce(FinalOutput,mValues)
    print("Final Output is : ", rValues)

if __name__ == "__main__":
    main()

```

5. Write a program which contains filter(), map() and reduce() in it. Python application which contains one list of numbers. List contains the numbers which are accepted from user. Filter should filter out all prime numbers. Map function will multiply each number by 2. Reduce will return Maximum number from that numbers. (You can also use normal functions instead of lambda functions).

Input List = [2, 70 , 11, 10, 17, 23, 31, 77]
 List after filter = [2, 11, 17, 23, 31]
 List after map = [4, 22, 34, 46, 62]
 Output of reduce = 62

Answer:

```

from functools import reduce

def PrimeNumbers(iValues):
    for i in range(2,iValues):
        if iValues%i==0:
            return False
    return True

Multiplication = lambda iValues : iValues * 2

```

```
def Max(iValue1,iValue2):
    if iValue1 < iValue2:
        return iValue2

def main():
    print("Enter the list of Numbers")
    Values = list(map(int,input().split()))

    fValues = list(filter(PrimeNumbers,Values))
    print("List after Filter : ", fValues)

    mValues = list(map(Multiplication, fValues))
    print("List after Map :", mValues)

    FinalOutput = reduce(Max,mValues)
    print("Final Output is : ", FinalOutput)

if __name__ == "__main__":
    main()
```